

ED-001 Ontology: Methodology and Research Gap

Emotion-Aware Decision Support for Critical Learning Situations

Methodology

ED-001 is built using a five-layer ontology architecture aligned with Control-Value Theory (Pekrun, 2006). The design methodology follows established semantic web standards (OWL DL, SWRL, SPARQL) and incorporates reuse of validated external ontologies (MFOEM, EFO, Cognitive Atlas, BFO, DOLCE). The ontology structure implements a formal causal chain: Learning Context → Emotion → Cognitive State → Decision Strategy → Learning Outcome. Each layer is defined through explicit class hierarchies, object properties with domain/range constraints, and datatype properties capturing measurable attributes (task difficulty, emotion intensity, cognitive load level). SWRL inference rules operationalize the causal relationships, enabling the Pellet or HermiT reasoner to derive high-level cognitive and decision states from low-level behavioral signals and emotional states. Machine learning integration is achieved through a translation layer that converts ML classifier outputs (emotion probabilities from behavioral signals: response speed, error rate, question switching) into OWL semantic assertions. The ontology is evaluated through three competency questions operationalized as SPARQL queries over a simulated knowledge graph of five learner profiles in examination scenarios, with 25 total instances representing diverse emotion-task-outcome combinations. Adaptive recommendations are generated via SWRL rules mapping detected emotions and inferred cognitive states to three recommendation categories (Stress Reduction, Engagement Boost, Scaffolding Support), with confidence scores derived from emotion intensity and rule match strength.

Research Gap

No existing ontology simultaneously models all five layers of the emotion-cognition-decision causal chain. Emotion ontologies (EMO, MFOEM, EFO) define what emotions are but do not model their downstream influence on cognition or decision-making. Cognitive ontologies (Cognitive Atlas) represent mental processes but lack emotional triggers or outcomes. Learning system ontologies (Protus 2.0) model learner state and adaptive content sequencing but do not incorporate emotional causation or affective appraisals. The literature reveals a five-part gap: (1) No ontology formally represents emotional states with causal properties; (2) No ontology models the downstream influence of emotion on cognitive load and reasoning quality; (3) No ontology connects cognitive state to classified decision strategies; (4) No ontology links decision strategy to predicted learning outcomes; (5) No ontology integrates ML-based emotion detection with OWL/SWRL semantic reasoning. This gap is particularly acute in intelligent tutoring systems, which evaluate learners on task difficulty and prior performance but remain entirely blind to the emotional states driving decision quality. ED-001 closes this gap by providing the first machine-interpretable ontology that operationalizes the complete emotion→cognition→decision→outcome chain, enabling adaptive systems to condition support not just on what a learner did wrong, but on the emotional and cognitive state that caused the error. The ontology's multi-discipline alignment framework (educational psychology, intelligent tutoring systems, clinical psychology, human-computer interaction, affective computing) enables cross-domain knowledge exchange without data transformation or bespoke connectors, transforming emotions from unobserved confounders into observable, reasoned-about variables in learning systems.